

Momentum Risk-Cap Options Agent

An autonomous Alpaca agent that turns price momentum into defined-risk options spreads.

PAPER TRADING ONLY

OFFICIAL MCP SERVER

423 OFFLINE TESTS

The order of operations is the logic

`src/agent.py::run_once` — one pass, one underlying, four stages, always this order.



- Exits run first and are deliberately not gated by the daily-loss breaker: an agent that is losing money must still be able to close.
- Reconciliation runs after the exits, so a stop that fills on this pass is a loss the breaker already sees when the entry side is asked for a decision — reconciling first made every self-inflicted loss arrive one pass late.
- The order book is read before positions on purpose: a fill landing between the two calls is double-counted rather than missed.

From a price move to a defined-risk structure

src/strategy.py::MomentumRiskCapStrategy.decide — defaults shown, all configurable.

MOMENTUM $\geq$ 2%



CHAIN 7-21 DTE



LIQUIDITY SCREEN



RANK BY STRIKE



SELL THE FAR LEG

- Momentum over the last 10 daily closes; act only at $|\text{momentum}| \geq 2\%$. Positive → call structure, negative → put structure. Otherwise hold.
- Candidates are ranked by distance from spot, tie-broken by distance from mid-window and then by symbol, so the pick never depends on the order the chain came back in.
- A farther-OTM leg of the same type and expiry is sold against the long leg — a debit vertical, sized so the width is nearest 3% of spot.
- It falls back to a naked long only if the chain genuinely offers nothing cheaper, and the log says which of the two happened.

Three ways out, decided before entry

`src/exits.py::ExitPolicy` — evaluated on every pass, ahead of everything else.

TAKE PROFIT

+75%

of cost basis

STOP LOSS

−50%

of cost basis

TIME STOP

1 DTE

win or lose, no expiry risk

- Percentage tests apply only to structures with a positive net debit basis; credit or zero-basis structures get the time stop and no invented percentage.
- The time stop is checked ahead of both P&L rules on purpose: assignment is the outcome it exists to prevent.

Nine gates, every one before an order is sent

WRITEUP.md §2 — one row each, enforced in src/strategy.py, src/exits.py, src/portfolio.py.

01	Per-trade budget	risk_pct 1% of equity per ticket
02	Portfolio budget	max_portfolio_risk_pct 5%, sized on what is left
03	In-flight risk	working unfilled orders charged from submission
04	Daily-loss breaker	max_daily_loss_pct 3% halts new entries
05	Contract cap	max_contracts 5 per order, hard ceiling
06	Liquidity screen	max_spread_pct 10% of the contract's own mid
07	Limit-only entries	opening orders are limits at the sized price
08	Naked-short guard	refuse a close that leaves an uncovered short
09	Stand-down on unknowns	one unreadable term stands the entry side down

Alpaca's official MCP server, pinned

src/mcp_client.py::AlpacaMCPClient — uvx alpaca-mcp-server==2.3.0 over stdio.

SERVER	TOOLS CALLED	ARGUMENT NAMES	PREFLIGHT
2.3.0 pinned, not 'latest'	8 incl. mleg option orders	25 checked vs inputSchema	PASS zero mismatches, no account

- The wire contract was read from upstream source, not guessed: qty as a string, calendar days spanning the trading days asked for, cursor-paged orders and activities.
- scripts/preflight_live.py handshakes the real server with placeholder keys and checks every tool and argument name this repo sends against the server's own inputSchema — without touching an account.
- A byte-compatible mock backs --dry, so the whole decision path runs with no keys and no network: 423 tests.

Every decision names its own numbers

Not a report generated afterwards — the reason string is what the sizing code produced.

```
buy_call_spread SPY contracts=5 momentum=7.43% max_loss=$270.00 open_risk=$693.00
```

...sized against the binding budget (per-trade, \$1000.00; per-trade cap 1.0% of \$100,000.00 = \$1000.00, portfolio headroom \$4307.00 after \$693.00 already at risk)...
sold SPY260912C00110900 against it: \$2.20 wide vs \$3.26 target, cutting cost from \$1.48 to a \$0.54 net debit and capping max loss at \$54.00 per spread... long leg at the offer \$1.48 (last \$1.46), short leg at the bid \$0.94 (last \$0.95).

- Every pass is appended to journal/decisions.jsonl — the no-trade passes included.
- So any P&L a judge sees on the account traces back to the sentence that caused it.

What has not happened yet

WRITEUP.md §4. Stated here for the same reason it is stated there.

The agent has never read a live quote.

Every number in this deck comes from the mock or from a unit test.

- The dedicated paper account the rules require was not yet created when this deck was rendered, so the \$100,000 starting balance is untested against a real account.
- First live action is a read-only connectivity pass — clock, account, chain — not an order.
- No P&L, equity curve or win rate appears anywhere in this deck, on purpose.